

Callahan D. Carvalho

+1 (213) 791-7670 | callahancarvalho@gmail.com | linkedin.com/in/callahan-carvalho | github.com/DominicCloud | Portfolio

Professional Summary

Game Designer and Software Engineer specializing in narrative-systems design—creating mechanics that serve dramatic intent and evoke emotional resonance. Approaches game design through understanding player psychology: how systems, feedback loops, and interactive choices influence behavior, emotion, and interpretation. Combines writer’s perspective on thematic cohesion with systems-thinking methodology, treating mechanics as narrative language and game elements as resources within interconnected systems. Proven track record developing digital games and tabletop prototypes that prioritize meaningful player experiences through holistic design—where UI, sound, art, level design, and mechanics exist in constant dialogue. Maintains strong technical implementation capabilities across full-stack development, 3D graphics programming, and AI systems. Currently pursuing Master’s degree in Computer Science with Game Development specialization at University of Southern California.

Technical Skills

Game Development & Design: Godot Engine (GDScript), Unity – C# (Learning), Unreal Engine 5 (Learning), Blender (3D Modeling & Animation), System-Based Design Methodology, Level Design, Gameplay Systems Architecture, Game Balancing, Player-Centric Design, Finite State Machines, Resource Management Systems, AI Programming, Procedural Generation

Programming Languages & Frameworks: C++ (Advanced), Python, C, Java, GDScript, JavaScript, C#, Django, Flask, React, Next.js, Node.js

Graphics & Rendering: Ray Tracing, Software Rasterization, GPU Programming, Shader Development, Graphics Pipeline Architecture, Procedural Content Generation, Image Processing, 3D Rendering Algorithms

Web Development & Databases: Full-Stack Development, MongoDB, RESTful API Design, HTML5, CSS3, Responsive Design, Web3 Integration

Design Tools & Software: Adobe Creative Suite (Premiere Pro, Photoshop, Illustrator, InDesign), Inkscape (Vector Graphics), Libresprite (Pixel Art), Blender (3D Modeling & Animation), UI/UX Design, Brand Identity Development

Emerging Technologies: Blockchain Architecture, Smart Contract Development (Solidity), Decentralized Applications (dApps), Machine Learning (Scikit-learn), Linear Regression, Feature Engineering

Professional Competencies: Project Management, Cross-Functional Team Leadership, Iterative Development Methodologies, User Experience Design, Technical Documentation, Technical Support, Customer Service, Agile Workflows, Version Control (Git)

Education

University of Southern California (GPA: 3.85 / 4.0)

Master of Science in Computer Science – Game Development

Los Angeles, CA

Aug 2025 – May 2027

– Courses: Analysis of Algorithms, 3D Graphics and Rendering, Game Design Workshop, Interactive Media and Games

University of Mumbai (CGPA: 9.44 / 10.0)

Bachelor of Engineering in Computer Engineering

Mumbai, India

Jan 2021 – May 2025

St. Xavier’s College (Percentage: 94.00%)

Higher Secondary Education – Science Stream

Mumbai, India

Jan 2019 – May 2021

Don Bosco High School (Percentage: 91.20%)

Secondary School Certificate (SSC)

Mumbai, India

Jan 2009 – May 2019

Relevant Graduate Coursework

3D Graphics & Rendering

- Architected and implemented a comprehensive ray-tracer from scratch in C++, incorporating advanced reflection and refraction algorithms to simulate realistic light behavior and material interactions
- Engineered a complete software rasterizer without hardware acceleration, implementing the full graphics pipeline from vertex processing through fragment shading
- Developed procedural content generation systems utilizing Perlin and Simplex noise algorithms, and constructed a GPU-driven rendering pipeline leveraging modern graphics APIs

Game Design Workshop

- Conceptualized and prototyped four complete tabletop games from initial concept through iterative playtesting, applying player-centric design methodologies
- Led design team as Lead Designer for animal-themed social deduction game, orchestrating mechanics architecture and narrative framework aligned with core experience objectives

Professional Experience

Technical Support Specialist | ITS Customer Support, USC

Technical Support, Troubleshooting, Customer Service, Windows, macOS,

Los Angeles, CA

Oct 2025 – Present

- *Linux* Deliver comprehensive technical support to university community, troubleshooting complex hardware and software issues across Windows, macOS, and Linux platforms
- Diagnose and resolve technical challenges related to enterprise systems, learning management platforms, and network infrastructure while maintaining detailed support documentation
- Translate complex technical concepts into accessible language for diverse audiences, ensuring efficient problem resolution and high customer satisfaction

Lead Game Designer | USC Game Design Workshop

Tabletop Game Design, Social Deduction, Playtesting, Player-Centric De-

Los Angeles, CA

Nov 2025 – Dec 2025

- *sign* Directed cross-functional design team in creating player-centric animal-themed social deduction game, managing project from conceptualization through final prototype delivery
- Architected core game mechanics and narrative framework aligned with targeted emotional engagement and strategic depth objectives
- Conducted 15+ iterative playtesting sessions over 4-week development cycle, systematically refining social deduction mechanics through data-driven design decisions

Graphic & UI Designer | EvaluAItor (Internship)

UI Component Libraries, Brand Design, Adobe Creative Suite, Stakeholder

Remote

Mar 2024 – Aug 2024

- *Collaboration* Designed comprehensive UI component library and established cohesive brand color palette serving as foundation for all visual communications
- Produced daily visual content across social media platforms and official website, maintaining consistent brand identity and user engagement
- Collaborated with development team through iterative design sprints, implementing stakeholder feedback to refine interface aesthetics and functionality

Technical Head | Student Council, University of Mumbai

Event Management, Technical Infrastructure, Cross-Functional Leadership,

Mumbai, India

Aug 2023 – Apr 2024

- *Project Coordination* Managed technical infrastructure for three major institutional events: Mosaic (cultural festival), IRIS (technical symposium), and Ignitra (entrepreneurship summit)
- Coordinated cross-departmental teams and managed concurrent project timelines, successfully delivering all technical requirements on schedule

Graphics Head Student Council, University of Mumbai <i>Creative Direction, Visual Design, Brand Identity, Adobe Creative Suite</i> – Led creative direction and visual identity across all three major institutional events, maintaining consistent branding and design standards – Produced event collateral, stage design, and promotional materials, overseeing all graphic output from concept through final delivery	Mumbai, India Aug 2023 – Apr 2024
Head Curator TEDxSFIT: Time's Canvas, University of Mumbai <i>Content Curation, Speaker Selection, Presentation Coaching, Event Planning</i> – Curated speaker content and developed talk narratives aligned with 'Time's Canvas' theme while maintaining TEDx quality standards – Led curation team through speaker selection, content development, and presentation coaching processes to deliver cohesive event experience	Mumbai, India Jun 2023 – Oct 2023
Head of Design IRIS Magazine, University of Mumbai <i>Visual Design, Team Leadership, Magazine Layout, Cover Art</i> – Directed design team for annual institutional magazine, overseeing visual strategy and creative execution – Designed distinctive cover art establishing visual identity for publication and contributed to overall layout and content presentation	Mumbai, India Dec 2022 – Jan 2023

Game Development Portfolio

Digital Game Projects

Bardo – 2D Narrative Platformer Systems Designer, Programmer <i>FSM Architecture, Resource Management, Level Design, Godot Engine, GDScript, Pixel Art</i> – Architected robust Finite State Machine managing player movement states, animation transitions, and input handling with seamless state synchronization – Engineered comprehensive resource management system tracking health, humanity meter, and currency with real-time UI integration and visual feedback – Developed memory-optimized level manager implementing two-room buffer architecture, enabling instant loading transitions by caching adjacent area data – Designed complete first area featuring strategic trap placement, collectible distribution, and interconnected pathway networks encouraging exploration – Reimagined traditional obstacle design to accommodate death-as-progress mechanic, implementing Humanity Meter corruption system to maintain challenge balance	Los Angeles, CA Mar 2025 – Present
ChopstyX – Turn-Based RPG Producer, Programmer <i>Turn-Based Combat, Team Leadership, RPG Systems, Godot Engine, GDScript, Pixel Art</i> – Led three-person team through complete 48-hour Global Game Jam development cycle, managing scope, task allocation, and milestone delivery under extreme time constraints – Coordinated cross-disciplinary collaboration between programming, art, and design to deliver polished playable experience within jam timeframe – Adapted classic Chopsticks ruleset into turn-based combat system with RPG progression mechanics and special ability integration – Received Best Technical Design Award recognizing strong technical execution and effective team collaboration	Los Angeles, CA Jan 2026
Tower Defense Strategy Game Programmer, Designer <i>AI Programming, Pathfinding, Economy Balancing, Systems Design, Godot Engine, GDScript, Blender</i> – Implemented core systems including resource economy, pathfinding algorithms, tower placement validation, and automated turret AI with priority targeting – Balanced gameplay economy through iterative testing cycles, creating meaningful strategic choices while implementing performance-based rating system encouraging mastery	Mumbai, India Dec 2024

- | | |
|---|---------------|
| Rocket Boost – 3D Physics Platformer Programmer, Designer
<i>3D Development, Physics Systems, Scene Management, Level Design,</i>
<i>Godot Engine, GDScript, Blender</i> | Mumbai, India |
| <ul style="list-style-type: none"> – Designed five progressive levels with escalating difficulty curves, implementing precision physics-based controls requiring skill mastery – Developed scene management architecture and created diverse obstacle sets testing spatial awareness and strategic planning | Dec 2024 |
| The Firing Knight – 2D Pixel Platformer Programmer, Designer
<i>Enemy AI, Combat Systems, Animation, Tilemap Design, Godot Engine,</i>
<i>GDScript, Pixel Art</i> | Mumbai, India |
| <ul style="list-style-type: none"> – Designed gameplay loop integrating platforming precision with tactical combat, implementing enemy AI with patrol patterns and detection systems – Created level architecture using tilemap systems with strategic collectible placement encouraging exploration and rewarding skilled play | Jun 2024 |

Digital Prototypes

- | | |
|--|-----------------|
| Disc Runner – High-Precision Platformer Designer, Programmer
<i>Movement Mechanics, Camera Systems, Physics Design, Level Design,</i>
<i>Godot Engine</i> | Los Angeles, CA |
| <ul style="list-style-type: none"> – Engineered dynamic room-to-room camera system with intelligent framing logic automatically adjusting zoom levels and offsets for rooms of any dimension – Implemented seamless viewport-boundary transitions using predictive helper functions to maintain player momentum across room changes while optimizing off-screen asset culling | Jan 2026 |
| Pong-O – Competitive Arcade Game Designer, Programmer
<i>PvP Design, Score Systems, Circular Arena Mechanics, Risk-Reward Bal-</i>
<i>ance, Godot Engine</i> | Los Angeles, CA |
| <ul style="list-style-type: none"> – Designed rally multiplier system rewarding sustained skillful play while penalizing rally-breaking actions at 50x current multiplier, reinforcing high-risk decision-making – Implemented dynamic shrink system reducing paddle size over time while increasing base multiplier, naturally escalating difficulty and score potential with optional accessibility toggle | Jan 2026 |
| Snake Catcher – Idle Progression Game Designer, Programmer
<i>Idle Mechanics, Procedural Animation, Inverse Kinematics, Memory Opti-</i>
<i>mization, Godot Engine</i> | Los Angeles, CA |
| <ul style="list-style-type: none"> – Engineered procedural snake movement using Inverse Kinematics with joint-by-joint response system, enabling fluid organic motion without baked animations – Optimized memory efficiency through object pooling—despawned snakes reposition and recolor rather than destroy/recreate, enabling higher snake counts without performance cost | Jan 2026 |

Tabletop Game Prototypes

- | | |
|--|-----------------|
| Voidrunners – 4-Player Cooperative Space Adventure Designer
<i>Cooperative Mechanics, Resource Management, Risk-Reward Systems</i> | Los Angeles, CA |
| <ul style="list-style-type: none"> – Designed collaborative mechanics emphasizing resource sharing and strategic bumping to assist teammate movement – Implemented risk-reward system requiring gravitational trajectory calculations to optimize fuel conservation | Sep 2025 |
| Toy Box Brawl – 1v1 Territory Control Game Designer
<i>Competitive Design, Territory Control, Card Mechanics, Resource Manage-</i>
<i>ment</i> | Los Angeles, CA |
| <ul style="list-style-type: none"> – Balanced card-based resource management integrating action point allocation with strategic card deployment – Designed asymmetric risk-reward territory system where aggressive positioning yields higher Victory Points at increased tactical vulnerability | Oct 2025 |

Quit Lion! – Multilateral Bluffing Race Game | Designer

Los Angeles, CA

Nov 2025 – Dec 2025

Social Deduction, Bluffing Mechanics, Multilateral Competition

- Innovated traditional bluffing mechanics by embedding deception into core movement rather than supplementary systems
- Designed multilateral gameplay requiring simultaneous tracking of multiple opponents' potential bluffs while managing personal deception strategy

Madjack – Blackjack with Meta Progression | Designer

Mumbai, India

Dec 2025

Meta Progression, Card Game Redesign, Strategic Depth

- Redesigned core Blackjack mechanics incorporating powerup systems and special abilities transcending traditional probability calculations
- Implemented meta progression architecture carrying strategic advantages across rounds, rewarding long-term planning and creating narrative engagement

Graphics & Rendering Projects

Procedural Snake Animation System | Graphics Programmer

Los Angeles, CA

Nov 2025 – Dec 2025

Procedural Generation, Unity, C#, Perlin Noise, Custom Shaders, CCDIK,

Terrain Generation, Poisson Disk Sampling

- Developed procedural terrain generation system using Perlin noise heightmaps with customizable octaves, lacunarity, and noise scale parameters for real-time environment creation
- Engineered TerrainBuilder component with Level of Detail controls and height curve mapping for optimized mesh rendering and terrain steepness control
- Implemented environmental object distribution using Poisson Disk Sampling for realistic placement of trees and rocks across generated terrain

Software Engineering Projects

Shuttle4Space – Decentralized Storage Platform | Lead Developer

Mumbai, India

Jul 2024 – Apr 2025

Blockchain, Smart Contracts, Web3, UI/UX Design, Published Research

- Architected decentralized peer-to-peer disk space rental solution leveraging blockchain technology for secure, transparent, tamper-proof transactions
- Implemented smart contracts automating rental agreements, payment processing, and dispute resolution within trustless system architecture
- Developed comprehensive user interface integrating blockchain wallet connectivity and real-time transaction monitoring for providers and renters

Movie Rating Prediction System | Developer

Mumbai, India

Jul 2024 – Oct 2024

Machine Learning, Python, Scikit-learn, Linear Regression, Feature Engi-

neering

- Developed machine learning model utilizing Linear Regression to predict user movie ratings (1.0-5.0 scale) based on demographic and preference data
- Engineered feature extraction pipeline incorporating user demographics and viewing history for accurate prediction modeling
- Conducted comprehensive model validation using real-world datasets, achieving high prediction accuracy through iterative optimization

Tether by CommunityConnect – Charity Platform | Full-Stack Developer

Mumbai, India

Jul 2023 – Dec 2023

Django, React, MongoDB, Payment Processing, Full-Stack Development

- Built centralized platform enabling charities, NGOs, and fundraisers to host campaigns with comprehensive management tools
- Implemented secure payment processing, donation tracking, and user-friendly interface simplifying charitable giving while ensuring transparency

Walmart Supply Chain Management System | Developer

Mumbai, India

AI/ML, LSTM, Random Forest, Streamlit, Python, Supply Chain Optimization, Demand Forecasting, Data Visualization

Apr 2025

- Developed AI-powered supply chain management system implementing LSTM neural networks for 7-day demand forecasting and Random Forest models for inventory optimization
- Engineered comprehensive Streamlit dashboard featuring real-time data visualization, supplier performance analysis, and stock status monitoring across product categories
- Implemented inventory optimization system analyzing current stock, min/max thresholds, unit pricing, and lead times to calculate optimal stock levels

3D Modeling & Digital Art

Urban Environment Model | 3D Artist

Mumbai, India

Environmental Modeling, Architecture, Blender

Jun 2023

- Designed detailed urban scene featuring architectural clusters with adjacent street infrastructure, demonstrating understanding of environmental modeling

Everyday Objects Collection | 3D Artist

Mumbai, India

Hard-Surface Modeling, Organic Modeling, Blender

Jun 2023

- Modeled diverse object collection (mug, trumpet, pencil, apple, sunglasses, headphones, piggy bank, heel shoe) demonstrating versatility across hard-surface and organic modeling techniques

Publications & Research

Shuttle4Space: Space Renting System using Blockchain

May 2025

Journal of Technology, Issue 5 (Paper ID: JOT-7225)

Authored research paper presenting comprehensive design and implementation of decentralized peer-to-peer disk space storage and rental solution leveraging blockchain technology for secure, transparent, tamper-proof transactions in distributed storage systems

Certifications & Awards

Prakalp Project Competition – First Place

Jan 2025

USC Global Game Jam '26 – Best Technical Design Award

Jan 2026

Microsoft Learn – Azure AI Fundamentals

Dec 2023

Scaler Academy – JavaScript Programming

Jan 2023

Bolt IoT – Python Programming

Jan 2022

IEEE – Git & GitHub Version Control

Jan 2022

Languages

English: Native/Fluent

Hindi: Native/Fluent